Reversal in Cryptocurrency Returns

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May 3, 2026

Abstract

We document significant reversal in cryptocurrency returns at 8- and 10-week horizons, concentrated in midsize, relatively volatile assets. Using a panel of 70 USDT-quoted tokens on Binance from January 2021 through March 2026, we show that a contrarian strategy of buying past losers and selling past winners, by forming Jegadeesh-Titman (1993) calendar-time overlapping portfolios, earns a 39.6% annualized return (Sharpe 0.96, Newey-West t = 2.10). Reversal is stronger among high-volatility assets, generating a Sharpe ratio of 1.37 (t = 3.19), and is strengthened outside of the largest assets, generating a Sharpe ratio of 1.69 (t = 3.80). The effect is robust across tercile, quintile, and decile sorts; skip period variants; inverse-volatility weighting; and temporal subsamples. A circular block bootstrap with 10,000 replications corroborates the high-volatility result nonparametrically with 95% of Sharpe ratios above 0.67, and the high-versus-low volatility gap positive in 94% of replications. Several economic mechanisms to rationalize these findings are discussed, including the tendency of treasury managers to sell into upswings, and Nagel's (2012) theory that reversal compensates liquidity providers.

1 Introduction

Momentum — the tendency of past winners to continue outperforming past losers — is a robust finding in empirical asset pricing and is a widely used signal by active investors. Jegadeesh and Titman (1993) documented that equity momentum strategies earn approximately 1% per month at 3–12 month horizons, a finding that has been replicated across dozens of markets and asset classes (Asness, Moskowitz, and Pedersen 2013). Equally important is the finding that momentum profits eventually *reverse*: Jegadeesh and Titman (2001) showed that the winners-minus-losers portfolio earns negative returns at 13–60 month horizons, and De Bondt and Thaler (1985) documented that 3–5 year losers outperform winners by approximately 8% per year.

Shorter-horizon reversal effects have also been documented in equities but operate on a different timescale and are typically attributed to microstructure frictions rather than something like overreaction. Jegadeesh (1990) found a roughly 2% monthly reversal in US equities at the 1-month formation horizon, and Lehmann (1990) documented weekly reversal attributable largely to bid-ask bounce and inventory pressure. These short-horizon effects are distinct from the intermediate-horizon (6–12 week) reversal we document in crypto, which sits at the horizon where equity markets instead exhibit momentum.

In cryptocurrency markets, Liu, Tsyvinski, and Wu (2022) established that momentum operates at 1 to 4 week horizons for value weighted portfolios. While Liu et al. study a larger cross section, we replicate Liu's momentum result on our sample. However, the reversal and momentum results

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appear to be distinct phenomena playing out among slightly different assets. The reversal signal is equal weighted, strengthened when sorting by high volatility, and strengthened among our mid-capitalization assets.[1]

To our knowledge, our findings present the strongest evidence to date for intermediate-horizon reversal in cryptocurrency returns. The effects we find are amplified in volatile mid-market capitalization assets, and in assets that don't have fixed or programmatic supply like Bitcoin. Turnover and transaction costs are tractable so the signal is actionable. The findings suggest several potential economic explanations for reversal, including treasury management by foundations and cryptocurrency project entities, and Nagel's (2012) compensation for liquidity argument, which are discussed in targeted tests and the concluding remarks.

In order to isolate the strategy, we characterize cross-sectional reversal in crypto returns across a grid of formation ($J = 4$ to 26 weeks) and holding ($K = 1, 2, 4$ weeks) periods. We find that the contrarian Losers-Minus-Winners (LMW) portfolio peaks at $J = 8$ weeks with a Sharpe ratio near 1.0. We show through conditional double sorts that reversal is six times stronger for high-volatility assets. The high-vol LMW Sharpe of 1.36 is insensitive to dropping LUNA, FTT, and ARDR, so the result is not driven by event-specific outliers. We run extensive robustness checks across sort procedures, return definitions, weighting schemes, and subsamples.

Over the past decade, researchers have recommended higher t-statistic thresholds to account for the accumulation of multiple hypothesis tests on the traditional datasets available to financial economists (Harvey, Liu, and Zhu 2016). Our setting differs in an important respect: we are not proposing a new anomaly but testing a well-documented one — intermediate-horizon reversal, known since De Bondt and Thaler (1985) and Jegadeesh (1990) — on a dataset from a market that had not yet been established when the original relationship was estimated. Our strongest specifications clear the recommended threshold regardless: the high-volatility conditional sort produces t = 3.19 and the mega-cap exclusion reaches t = 3.80.

The remainder of the paper is organized as follows. Section 2 describes the data and methodology. Section 3 presents the main cross-sectional reversal results including the conditional double sorts that identify volatility as a key driver. Section 4 presents robustness checks and Section 5 concludes. A battery of robustness tests and technical details are relegated to the Appendix.

2 Data and Methodology

2.1 Data

Our sample consists of 70 USDT-quoted cryptocurrency pairs traded on Binance, observed daily from January 1, 2021 through February 28, 2026 (1,885 trading days).

Mid-price reconstruction. Binance trade data report each execution's price and an aggressor flag (maker/taker direction) but do not publish top-of-book quotes historically. We reconstruct midprices using the Glosten-Harris (1988) price impact framework. Specifically, we estimate the two-parameter specification of Foucault, Pagano, and Roell (2013, equation 5.8), which decomposes the observed transaction price change into a permanent information component and a transitory bid-ask-spread component. The parameters are estimated via non-negative least squares (NNLS) to enforce a non-negative implied half-spread. Each trade's midprice is then reconstructed by adjusting

[1] Liu universe grows from 109 → 1,583 over 2014-2018. Our panel: 70 USDT-Binance pairs from 2021-2026. Liu reports momentum is statistically significant in above-median-size and insignificant in below-median-size, consistent with our VW J=2 result (t = 1.95*) but insignificant EW at J=2.

the trade price by half of the estimated spread in the direction opposite the aggressor flag reported by Binance (subtracting for buys, adding for sells).

The regression parameters are estimated on rolling windows of approximately 1,000 trades rather than fixed calendar intervals. Trade arrival rates rise during periods of market stress, so the window fills faster and the spread estimate updates more frequently when conditions are volatile. A clock-based window, by contrast, would average over both calm and stressed subperiods within a single day. When order flow within a window is too one-sided to reliably identify both the buy and sell intercepts, the window is expanded beyond 1,000 trades until both sides have sufficient representation.

Daily sampling. For each day, we identify the closest trade at or before the 16:00 ET timestamp within a tolerance of 5 minutes (chosen to align with NYSE close and facilitate comparison to equity benchmarks). If no trade is found within that tolerance window, the observation is marked missing. This procedure yields a point-in-time daily midprice that is free of bid-ask bounce and consistent across assets. Daily prices are resampled to weekly frequency (Sunday close). The resulting panel spans 269 weeks across 70 assets. All symbols have at least 52 weeks of non-missing observations; assets enter and exit the eligible set as data becomes available, producing an unbalanced panel at the weekly level.

Panel quality. The resulting panel is 98.7% complete. Of the 131,950 total cells (1,885 days $\times$ 70 symbols), only 1,756 are missing (1.33%). Twenty-five of the 70 symbols have zero missing observations; the median symbol has 0.27% missing data with a maximum gap of 2 days. Three symbols exhibit notable gaps driven by market events: FTT (312-day gap following the FTX exchange collapse in November 2022), LUNA (19-day gap during the Terra/LUNA collapse in May 2022), and ARDR (8-day gap of unknown origin). There are no zero prices, negative prices, or stale price sequences in the data. Ninety-seven daily price changes exceed 50% in absolute value across 40 symbols; inspection confirms these reflect legitimate crypto volatility events rather than data errors.

2.2 Portfolio Construction

We follow the calendar-time overlapping portfolio approach of Jegadeesh and Titman (1993). Each week t, we compute the cumulative return of each asset i over the formation window:

$$s_{i,t} = \frac{P_{i,t-\text{skip}}}{P_{i,t-\text{skip}-J}} - 1,$$

where J is the formation period in weeks and skip $= 1$ week is the skip period that avoids microstructure contamination. Assets with a valid (non-missing) signal $s_{i,t}$ are sorted into N quantile groups. Our baseline uses $N = 5$ (quintiles), with Q1 containing the lowest past returns (losers) and Q5 containing the highest (winners).

The contrarian long-short portfolio is

$$r_t^{\text{LMW}} = r_t^{Q1} - r_t^{Q5},$$

where r_t^{Qk} is the equal-weighted return of quantile group k in week t. When the holding period $K > 1$, we implement the Jegadeesh-Titman (1993) overlapping portfolio approach: at week t the portfolio return averages across K active cohorts,

$$r_t^{\text{LMW}}(K) = \frac{1}{K} \sum_{j=1}^{K} r_t^{\text{LMW, cohort formed at } t-j}.$$

We test formation periods $J \in \{4, 6, 8, 10, 12, 16, 20, 26\}$ weeks and holding periods $K \in \{1, 2, 4\}$ weeks.

2.3 Test Statistics

All t-statistics use Newey-West (1987) heteroskedasticity and autocorrelation consistent (HAC) standard errors with a Bartlett kernel and 6 lags, appropriate for the weekly frequency and the serial correlation induced by overlapping portfolio construction. Annualized Sharpe ratios for the zero-cost LMW portfolio are computed as $\text{SR} = (\bar{r}/\hat{\sigma}_r) \times \sqrt{52.18}$; because LMW is a zero-investment long-short portfolio, the return is already an excess return by construction and the risk-free rate is not subtracted.

3 Reversal Results

3.1 The J/K Grid

Table 1 reports the annualized LMW return, Sharpe ratio, and Newey-West t-statistic for each formation-holding period combination.

Table 1: LMW Portfolio Performance Across Formation (J) and Holding (K) Periods

J (wk)	K=1 Ann.	K=1 SR	K=1 t	K=2 Ann.	K=2 SR	K=2 t	K=4 Ann.	K=4 SR	K=4 t
4	−1.7%	−0.04	−0.09	6.1%	0.16	0.39	21.4%	0.63	1.38
6	16.4%	0.35	0.75	23.1%	0.58	1.29	**38.8%**	**1.11**	**2.29****
8	**39.5%**	**0.96**	**2.10****	**35.9%**	**0.88**	**1.85***	**34.9%**	**0.98**	**2.01****
10	30.3%	0.76	1.61	27.1%	0.68	1.44	28.3%	0.76	1.61
12	27.2%	0.65	1.51	25.6%	0.62	1.42	23.6%	0.64	1.42
16	20.1%	0.49	1.12	12.9%	0.33	0.78	8.1%	0.22	0.56
20	5.9%	0.14	0.37	3.2%	0.08	0.20	5.3%	0.14	0.37
26	1.7%	0.04	0.10	2.3%	0.06	0.14	5.8%	0.16	0.39

The effect is economically and statistically significant across $J = 6$–12 zone for all holding periods, with peak Sharpe ratios around 1.0 and t-statistics exceeding 1.85 in multiple (J, K) cells. The headline $J = 8$, $K = 1$ cell (SR = 0.96, $t = 2.10^{**}$) is not an isolated result: the two overlapping-holding variants at the same formation horizon ($K = 2$ and $K = 4$) also deliver significant reversal returns ($t = 1.85^{*}$ and 2.01^{**} respectively).

The dependence on K varies meaningfully with J. For short formation windows ($J = 4, 6$), longer holding periods substantially improve performance — $J = 6$ rises from SR = 0.35 at $K = 1$ to SR = 1.11 at $K = 4$ ($t = 2.29^{**}$), the highest individual Sharpe in the grid. For medium formation ($J = 8$–12), results are approximately K-invariant, indicating this is the peak-reversal zone where the signal is fully developed. For long formation ($J \geq 16$), $K = 1$ slightly dominates.

The effect decays monotonically to zero for $J \geq 20$ across all K, corroborating the finding that reversal is concentrated in the 6–12 week window and does not extend to the quarter-plus horizons found in the long-term equity reversal literature (De Bondt and Thaler, 1985).

3.2 Quintile Returns and Monotonicity

Table 2 reports annualized returns for each quintile at $K = 1$, showing the source of the LMW spread. Each cell reports the annualized return (top) and Newey-West t-statistic against H_0: mean

return = 0 (bottom, in parentheses). The final column reports the LMW (Q1 − Q5) spread return and t-statistic.

Table 2: Quintile Annualized Returns by Formation Period ($K = 1$)

J (wk)	Q1 (Losers)	Q2	Q3	Q4	Q5 (Winners)	LMW (Q1−Q5)
4	20.9% (0.53)	46.2% (1.05)	30.5% (0.69)	22.6% (0.55)	22.6% (0.56)	−1.7% (−0.09)
6	20.2% (0.53)	26.8% (0.65)	15.3% (0.38)	5.3% (0.14)	3.8% (0.10)	16.4% (0.75)
8	**24.9**% (0.61)	31.6% (0.80)	33.2% (0.78)	−4.6% (−0.12)	**−14.7**% (−0.43)	**39.6**% (2.10**)
10	19.4% (0.47)	24.8% (0.60)	10.4% (0.28)	−10.0% (−0.29)	−10.9% (−0.31)	30.3% (1.60)
12	17.4% (0.42)	19.9% (0.50)	−1.7% (−0.05)	−12.6% (−0.37)	−9.9% (−0.29)	27.2% (1.51)
16	9.1% (0.22)	−6.8% (−0.17)	0.7% (0.02)	−14.9% (−0.40)	−10.9% (−0.32)	20.1% (1.12)

At $J = 8$, the LMW spread is driven by both sides: Q1 (past losers) earns +24.9% and Q5 (past winners) earns −14.7%, though neither individual quintile return is significantly different from zero at conventional levels. The LMW spread itself is significant ($t = 2.10$) because the two errors are pooled into a long-short construction that removes common market noise. The quintile ordering is approximately reverse-monotonic at $J = 8$–12 (Q1 > Q2 > ⋯ > Q5), confirming that the effect is not driven solely by extreme quintiles.

3.3 Sorting on Volatility

There are several economic motivations for evaluating reversal among the relatively volatile subset of the assets in each sort. The Risk-Premium PCA analysis in Kiefer and Nowotny (2026) finds portfolios that do well when sorting on high means and high volatility. Mechanically, RP-PCA augments the sample covariance matrix with the outer product of the mean return vector — $\Sigma + \gamma\mu\mu^\top$ — so the resulting factors concentrate on directions that simultaneously explain return variance and have large expected returns. For some of the mean and covariance estimation window configurations, these portfolios may intersect with the reversal portfolios particularly if the reversal (mean-sorted) portfolios are also screened for volatility.

A complementary motivation from the equity literature comes from Nagel (2012), who argues that short-term reversal reflects compensation for liquidity provision: market makers who absorb selling pressure earn a premium that is larger for illiquid, volatile assets. The prediction is that reversal compensation scales with the shadow cost of liquidity provision, proxied by properties like volatility.

We independently sort assets into two groups by a conditioning variable, including trailing volatility, move extremity, and cross-sectional dispersion — and forming contrarian LMW portfolios within each group. These screens are designed to help connect the reversal signal to known phenomena, like the RP-PCA pattern and the Nagel channel, and ultimately refine its performance.

We condition on three variables, each measured at the portfolio formation date ($J = 8$ weeks): (i) trailing 13-week return volatility, (ii) absolute value of the formation-period return (move extremity), and (iii) cross-sectional return dispersion.

Table 3: Conditional Double Sort Results ($J = 8$ weeks)

Conditioning Variable	Group	LMW Ann. Return	Sharpe	NW t-stat
Volatility	Low	13.1%	0.30	0.57
Volatility	**High**	**81.8%**	**1.37**	**3.19*****
Move Extremity	Low	16.9%	0.41	1.02
Move Extremity	**High**	**46.0%**	**0.78**	**1.73***
CS Dispersion	Low	24.2%	0.46	1.06
CS Dispersion	**High**	**61.8%**	**1.17**	**2.51****

The high-volatility result (81.8% annualized, $t = 3.19$) is notable. Reversal among volatile assets is six times larger than among low-volatility assets. The same pattern holds for move extremity (consistent with Makarov and Schoar 2020, who document mean-reversion after extreme crypto price deviations) and cross-sectional dispersion. Circular and stationary block bootstraps corroborate the result: of 10,000 replications, 82% vol-high LMW Sharpe ratios exceed 1.0, with the joint high-vs-low gap distribution lying above zero in 93–94% of replications across block sizes from 4 to 12 weeks (Appendix D.3, Tables 16–18).

The sort findings are consistent with the Nagel (2012) liquidity-provision mechanism imported from equity microstructure. Testing additional Nagel-specific predictions requires additional data on LP positions and order flow, and we flag disentangling this story for future work. The findings may also interact with token selling and issuance policies, discussed in the treasury management section.

Another potential mechanism is the Scheinkman and Xiong 2003 sentiment-driven overreaction under heterogeneous beliefs. When short-sale constraints bind, optimists can push prices above fundamentals because pessimists cannot easily express their views. In crypto, shorting is available through perpetual futures but requires margin that scales with volatility, so the constraint is tighter for small, volatile tokens. This mechanism predicts asymmetric reversal: overpriced winners should revert more sharply than underpriced losers, since the upward distortion from constrained pessimists is the source of mispricing. However, table 2 shows the opposite pattern at J=8: Q5 (winners) earns -14.7% annualized while Q1 (losers) earns +24.9%, with the loser side contributing roughly 60% of the spread. The approximate symmetry, with a slight tilt toward the loser side, is more naturally explained by Nagel's liquidity-provision channel, where both buying losers and selling winners involve absorbing flow imbalances, than by a short-sale-constraint story that operates primarily on the winner side.

The vol conditioning variable is a 13-week trailing standard deviation, deliberately longer than the $J = 8$ formation horizon. We sweep vol windows $\in \{4, 8, 13, 20, 26\}$ weeks against $J \in \{6, 8, 10, 12\}$ and find that 13 weeks is the $J = 8$ optimum (high-vol LMW Sharpe 1.36 at vw = 13 vs. 1.23 at vw = 8 matched and 1.10 at vw = 4 short; long windows $\geq$ 20w degrade further). Window mismatch helps: at every J the best vol window is not equal to J, with shorter J's preferring shorter vw and $J = 8$ uniquely preferring vw = 13.[2] The independent double-sort construction is: at each rebalance we sort the cross-section into two vol halves first, then quintile-sort by formation signal *within* each half, and form the LMW spread (Q1 − Q5) inside the high-vol subset.

[2]This is broadly consistent with the Kiefer-Nowotny (2026) RP-PCA mechanism that a mean-estimation window distinct from the covariance window improves out-of-sample Sharpe ratio.

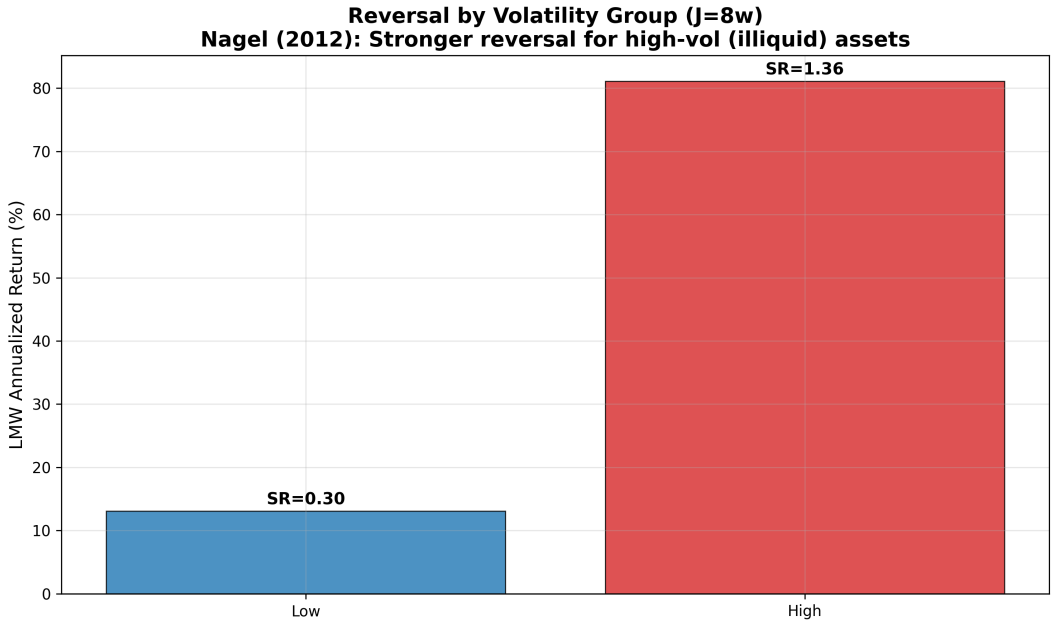

Figure 1: **Volatility double-sorted LMW cumulative returns ($J = 8$ weeks).** The figure compares the Sharpe ratios and mean returns for two LMW portfolios formed at $J = 8$ weeks: one within the high-volatility half of the cross-section and one within the low-volatility half. The high-vol portfolio delivers 81.8% annualized with Sharpe 1.37 ($t = 3.19$), versus 13.1% annualized and Sharpe 0.30 for the low-vol portfolio.

3.4 Relation to Liu et al. Momentum

Liu et al. document a momentum factor that helps explain the cross section of cryptocurrency returns in a multifactor model. Their momentum factor is based on value weighted portfolios formed on past portfolios returns over 1-4 weeks. We replicate that pattern in our panel, and show the reversal performance comes from a signal concentrated in a different set of assets.

When excluding the top market cap assets from the VW and equal weighted sorts, the short term momentum disappears but the 8-12 week reversal strengthens. Performance with the top 5 excluded is shown in 3 and is representative of excluding the top 4-8 by market capitalization.

While the Liu results and ours carve out a lifecycle that is compacted relative to equities, we have not established a within-crypto momentum-reversal lifecycle. In general the assets that exhibit momentum in the near term are not the ones that reverse over the J=8-12 horizons. One of the explanations for the lifecycle in equity is an overreaction-correction pattern, but this logic cannot be applied to the findings as they stand within crypto without testing something like whether, once corrected for sustained sell pressure from projects, tokens do exhibit the full lifecycle and the sell pressure kills the momentum.

3.5 Reversal is not the Market

We estimate the crypto market beta implied by our universe of assets

$$r_t^{\mathrm{LMW}} - r_f = \alpha + \beta(r_t^{\mathrm{MKT}} - r_f) + \varepsilon_t,$$

where $r_f = 3.5\%/52.18$ per week. We use two market benchmarks: an equal-weighted (EW) market return computed as the cross-sectional mean of all available asset returns each week, and a value-

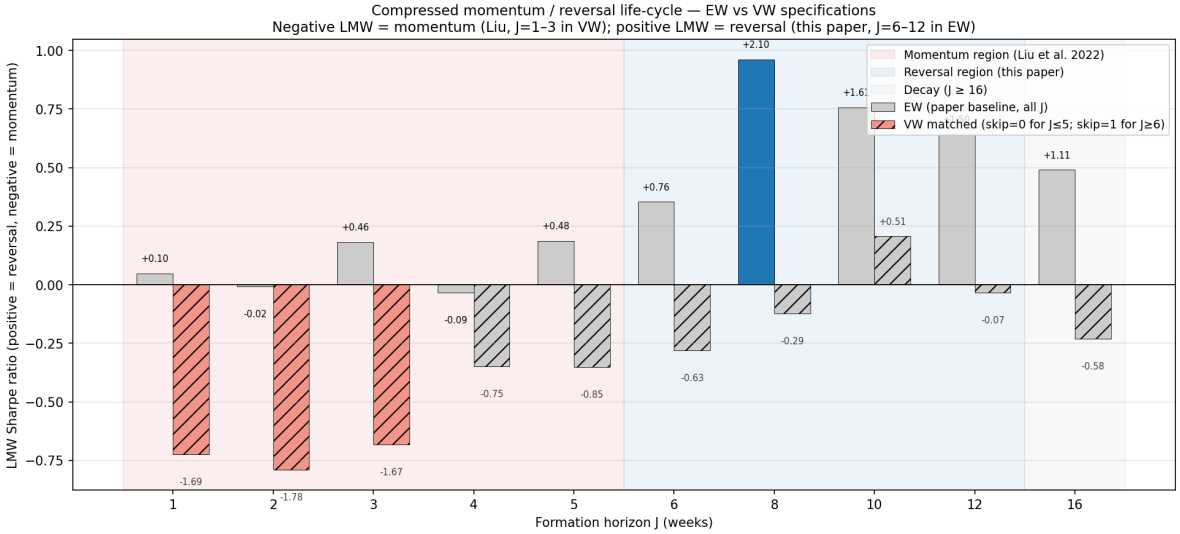

Figure 2: **Momentum-Reversal Lifecycles by Weighting**. Short horizon momentum is concentrated in larger, more liquid assets. Reversal is concentrated in smaller, higher volatility and less liquid assets. VW momentum is significant for $J = 1, 2, 3$ while EW reversal is significant over $J = 8, 10, 12$.

weighted (VW) market return using lagged market capitalizations (price times circulating supply from CoinGecko) as weights. The VW benchmark covers all 70 panel assets. The LMW portfolio is formed from the EW signal in each case. Alpha t-statistics use Newey-West standard errors with 6 lags.

Table 4: Crypto Market Alpha of LMW Portfolios

J (wk)	EW Alpha (ann.)	EW t-stat	VW Alpha (ann.)	VW t-stat	Beta (EW)	R^2
6	12.0%	0.55	12.8%	0.59	0.015	0.2%
8	**33.3**%	**1.83***	**33.7**%	**1.85***	0.043	2.2%
10	24.6%	1.33	24.4%	1.34	0.040	2.0%
12	21.5%	1.23	21.3%	1.24	0.042	2.0%
16	14.4%	0.84	14.4%	0.84	0.046	2.6%

Market betas are economically negligible (0.01–0.14); R^2 values are 2–3%. LMW returns are nearly orthogonal to the crypto market factor. At $J = 8$, alpha is 33–34% annualized ($t \sim 1.85$), significant at 10%. The EW and VW benchmarks yield virtually identical results. The precise OLS specification and the VW weighting construction are stated in Appendix B.

3.6 The Role of Treasury Management

We classify the 70 panel tokens into four categories based on how new and already-allocated supply enters the market. **Category 1** (PoW miner-only, fixed cap) covers chains where new supply enters only through mechanical block rewards and there is no significant foundation/lab allocation: BTC, LTC, BCH, ETC, DOGE, DASH, DCR, ZEC, RVN, DGB, SC, ZEN. **Category 2** (fixed cap with active treasury) covers tokens with a finite `max_supply` but a centralized issuer, foundation, or DAO that holds and sells already-allocated supply: BNB, FTT, XRP, AAVE, CRV, COMP, MKR,

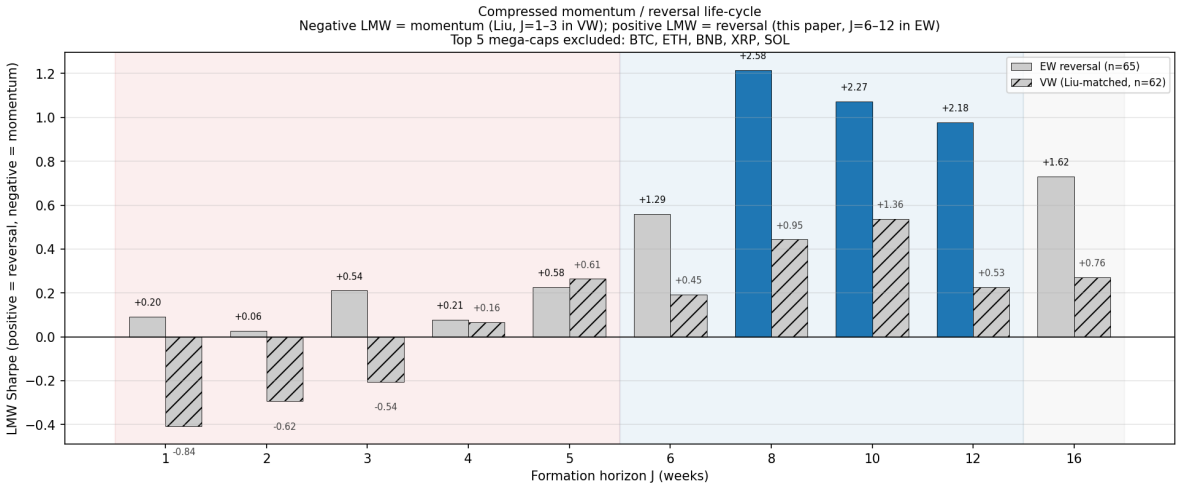

Figure 3: **Momentum-Reversal Lifecycles by Weighting - Excluding Largest** Value and equal weighted momentum and reversal portfolio Sharpe ratios excluding the top 5 cryptocurrencies by market capitalization.

UNI, LINK, MANA, and 17 others. **Category 3** (inflationary, passive foundation) covers protocols with uncapped programmatic emission and a documented passive treasury: ETH, ATOM, EGLD. **Category 4** (inflationary with active treasury) covers VC-backed L1/L2s and legacy foundation chains with documented recurring foundation-to-MM transfers: SOL, AVAX, ADA, ALGO, DOT, FIL, NEAR, XLM, XTZ, TRX, and 18 others.

Table 5: LMW J=8, skip= 1, $K = 1$ within supply-management categories. Sharpe ratio (NW t in parens). Sort dimension is quintile where $n_{\text{syms}} \geq 20$ and tercile otherwise; the high-vol/low-vol legs use a 2-bucket pre-sort by 13-week trailing volatility before the inner signal sort. Cat 3 (3 symbols: ETH, ATOM, EGLD) is too small for cross-sectional sorts.

Category	n_{syms} / sort	Unconditional	High-vol	Low-vol
1 PoW miner (fixed cap)	12 / q=3	−0.05 (−0.10)	+0.39 (+0.78)	−0.34 (−0.71)
2 Fixed cap, act. treasury	27 / q=5	**+1.06** (**+2.56****)	**+1.36** (**+3.34*****)	+0.56 (+1.36)
3 —	*N/A — N=3 symbols*			
4 Inflation, act. treasury	28 / q=5	**+0.97** (**+2.15****)	**+1.07** (**+2.47****)	+0.68 (+1.57)

Category 1 shows no cross-sectional reversal in either the unconditional cell ($SR = -0.05$, $t = -0.10$) or the high-vol cell ($SR = +0.39$, $t = +0.78$). Categories 2 and 4 — the two cohorts where a centralized entity is more likely to be actively monetizing already-allocated supply — both see large, statistically significant reversal. Cat 2 unconditional $SR = +1.06$ ($t = +2.56^{**}$) and high-vol $SR = +1.36$ ($t = +3.34^{***}$); Cat 4 unconditional $SR = +0.97$ ($t = +2.15^{**}$) and high-vol $SR = +1.07$ ($t = +2.47^{**}$).

4 Robustness

We present a battery of robustness checks to corroborate the reversal pattern's generalizeability.

4.1 Sort Granularity, Skips and Weighting

Table 6: LMW Performance by Number of Sort Groups ($K = 1$)

J (wk)	Terciles (N=3) SR	Quintiles (N=5) SR	Deciles (N=10) SR
6	0.48	0.35	0.05
8	**1.27**	**0.96**	**0.76**
10	**1.08**	0.76	0.70
12	**0.91**	0.65	0.56
16	0.34	0.49	0.32

At $J = 8$, the tercile sort yields Sharpe 1.27 with $t = 2.55^{**}$ (significant at 1%); the decile sort attenuates to $t = 1.56$ because groups become thin. The effect is consistent across sort granularities and is drawn out into J=10, 12 with the coarsest sort.

Table 7: LMW Sharpe Ratio by Skip Period ($N = 5$ quintiles, $K = 1$)

J (wk)	Skip = 0	Skip = 1 (baseline)	Skip = 2
8	0.66	**0.96**	0.50
10	**1.15** ($t = 2.30^{**}$)	0.76	0.49
12	0.80 ($t = 1.72^{*}$)	0.65	0.49

The reversal is robust to the skip period choice. Skip = 0 at $J = 10$ yields the highest t-statistic (2.30), suggesting the microstructure contamination traditionally avoided by the skip period is not a major concern in our weekly crypto data.

Table 8: Equal-Weight vs. Inverse-Volatility Weight ($K = 1$)

J (wk)	EW Ann.	EW SR	EW t	IV Ann.	IV SR	IV t
8	39.6%	0.96	**2.10**	28.4%	0.74	1.64
10	30.3%	0.76	1.60	27.1%	0.69	1.45
12	27.2%	0.65	1.51	20.2%	0.47	1.11

Inverse-volatility weighting reduces the reversal magnitude by 25–30%, consistent with the conditional-sort finding that high-volatility assets drive the effect.

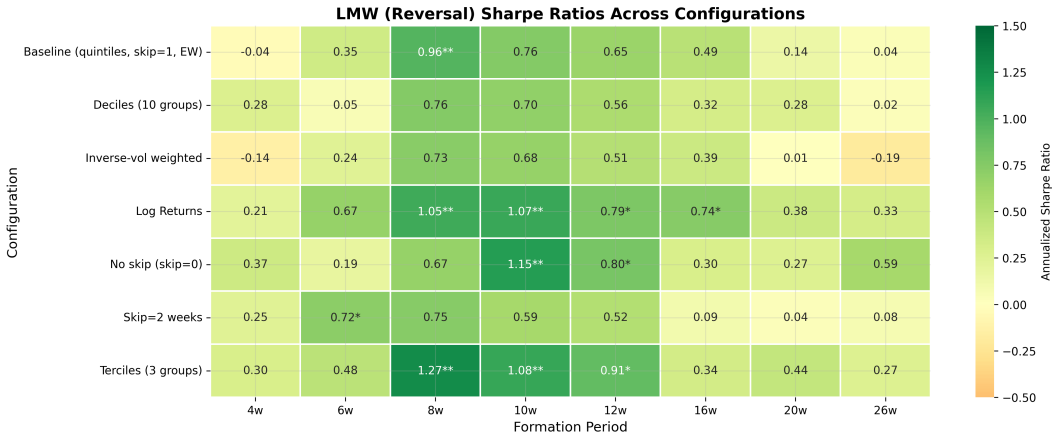

Figure 4: **LMW performance across the configuration robustness grid.** The figure plots LMW annualized performance against formation horizon J for the quintile baseline, tercile and decile sorts, skip = 0 and skip = 2 variants, log-return and inverse-volatility weighting variants, and value-weighted benchmarks. The J = 6–12 week zone is the locus where every variant performs, with tercile sorts and the skip = 0 specification delivering the highest Sharpes (1.27 and 1.15 respectively). Every curve decays toward zero for $J \geq 16$, none crosses zero significantly inside the J = 6–12 window, and no single variant drives the main $J = 8$ result.

4.2 Portfolio Turnover and Persistence

The $K = 1$ baseline implies weekly rebalancing of both the loser (Q1) and winner (Q5) quintiles, so realized transaction costs and the speed at which the cross-section shuffles through the extreme quantiles may matter for implementation. We measure portfolio turnover and asset persistence directly from the realized quintile membership at each weekly rebalance. One-sided weekly turnover is defined as $T_t^q = 0.5 \times |Q_t^q \triangle Q_{t-1}^q| / \overline{|Q^q|}$ (the fraction of names changed each week, with the denominator the average quintile size). Spell length is the number of consecutive weeks an asset stays in the same quintile across rebalances. Cohort survival at horizon k is the fraction of names entering Q at week t still in Q at week $t + k$, averaged across cohorts.

Table 9: Portfolio turnover and asset persistence.

Variant	Side	Turnover (mean / std)	Mean spell	Std spell	Max spell	k=1	k=4	k=8
Base LMW (J=8)	Q1 L	33.8% / 12.3%	2.94	2.95	31	66.2%	39.8%	19.2%
Base LMW (J=8)	Q5 W	25.7% / 9.6%	3.85	3.57	21	74.3%	46.8%	22.4%
V-H LMW (J=8)	Q1	**41.0% / 17.2%**	2.42	2.23	18	59.0%	27.5%	10.5%
V-H LMW (J=8)	Q5	**30.8% / 15.9%**	3.22	2.90	17	69.2%	38.3%	12.6%
V-L LMW (J=8)	Q1	38.8% / 16.6%	2.56	2.47	16	61.2%	34.0%	12.1%
V-L LMW (J=8)	Q5	34.0% / 16.4%	2.92	3.17	26	66.0%	40.2%	22.5%

Turnover is asymmetric across the long-short legs: Q1 refreshes 30–40% of names each week against 25–30% for Q5. Mean spell length is correspondingly shorter in Q1 (2.4–2.9 weeks) than in Q5 (3.2–3.9 weeks). Volatility conditioning increases turnover on both legs by roughly 5–7 percentage points. Despite the higher turnover, vol-high spell distributions have shorter right tails (max spell 17–18 weeks vs 21–31 weeks for baseline), reflecting that the very-persistent residency in baseline Q1 comes from low-vol perma-losers that vol-conditioning excludes. The implied holding period from turnover ($1/T$) is 2.4 weeks for vol-high Q1 and 3.0 weeks for baseline Q1.

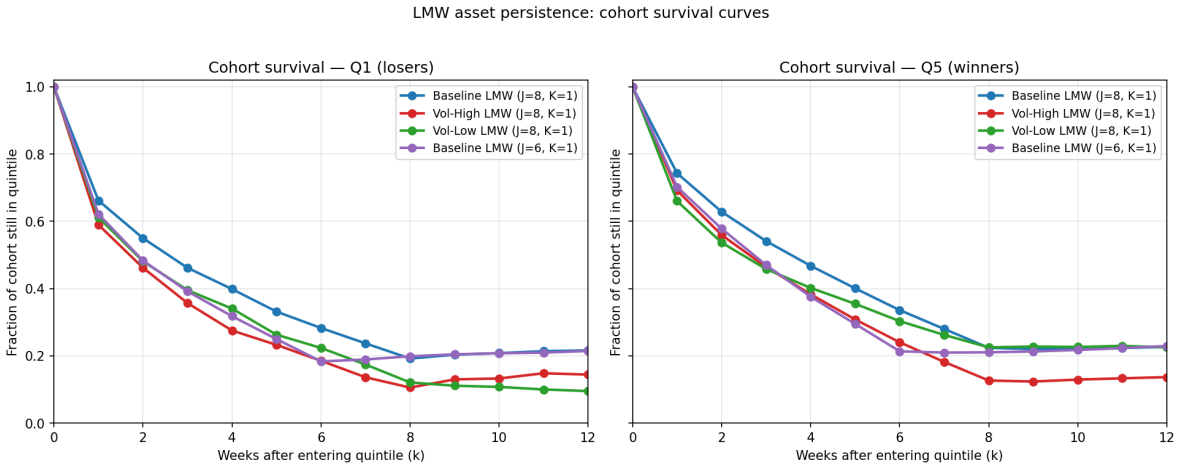

Figure 5: **Cohort survival in the extreme LMW quintiles.** For each cohort of names entering Q1 (losers, left) or Q5 (winners, right) at week t, the curves plot the fraction still in the same quintile k weeks later, averaged across cohorts. Survival decays roughly exponentially out to $k = 8$ weeks and then plateaus near 20% — a residual set of "regular residents" that occupy the extreme quintiles in roughly one in five weeks across the sample. The vol-high LMW (red) decays fastest from week 1 through week 8, consistent with the higher turnover; the plateau is also lower (∼10–15%), confirming that the high-vol cross-section is less dominated by perma-losers/perma-winners than the unconditional one.

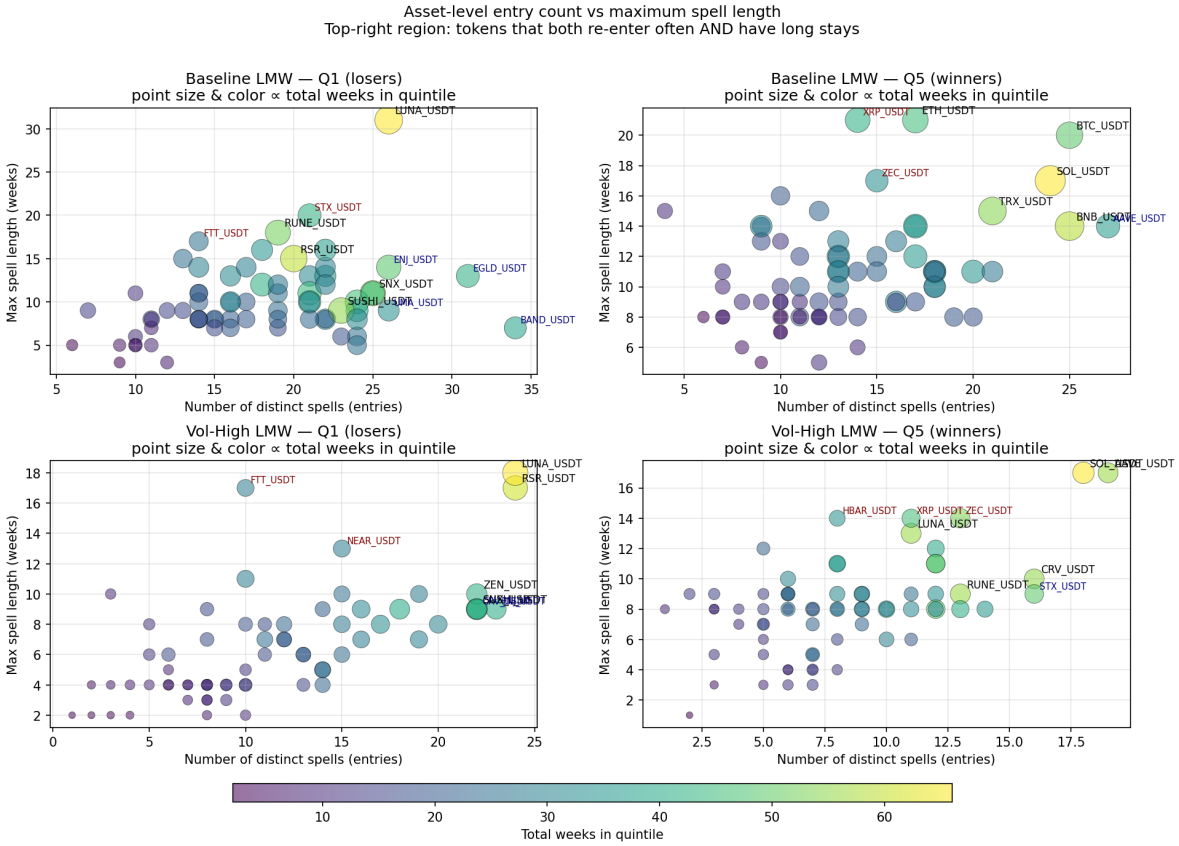

Figure 6: **Re-entry count vs maximum spell length per token.** Each point is one token; x-axis is the number of distinct spells (entries), y-axis is the longest single spell, point size and color encode total weeks in the quintile. The top-5 tokens by total weeks are labeled in black; outliers in either axis (95th percentile) are labeled in dark red (extreme single spell) or dark blue (frequent re-entry). The reversal results are robust to the exclusion of conspicuous outliers like LUNA.

Below we report additional transaction cost adjusted Sharpe rations, using a per-week transaction cost $= 2 \times (T^{Q1} + T^{Q5}) \times c_{\text{rt}}/2$, where c_{rt} is the round-trip cost in basis points and the leading 2 reflects that both legs of the LMW (long Q1, short Q5) trade. We sweep $c_{\text{rt}} \in \{10, 20, 50\}$ bps to bracket realistic crypto execution. These adjustments are on top of the bid-ask spread adjustments applied to the data using the Glostem-Harris procedure described in the data section.

Table 10: Transaction-cost-adjusted Sharpe ratio ($K = 1$)

Cost level	Base (J=8)	**VH (J=8)**	VL (J=8)	Base (J=6)
Sharpe ratio				
Gross	+0.96	**+1.36**	+0.30	+0.35
@ 10 bps	+0.82	**+1.27**	+0.15	+0.20
@ 20 bps	+0.67	**+1.15**	−0.02	+0.05
@ 50 bps	+0.22	**+0.77**	−0.54	−0.41
Annualized return				
Gross	+39.5%	**+81.1%**	+13.1%	+16.5%
@ 10 bps	+33.9%	**+75.6%**	+6.5%	+9.2%
@ 20 bps	+27.7%	**+68.1%**	−1.1%	+2.2%
@ 50 bps	+9.1%	**+45.6%**	−23.8%	−19.0%

The vol-high LMW survives 50 bps round-trip costs with Sharpe 0.77 and 45.6% net annualized return. The result is striking given that 50 bps is generous for institutional crypto execution and most mid-cap altcoin pairs trade at considerably lower spreads. Even at 20 bps the strategy clears Sharpe 1.0. The baseline LMW survives at 10–20 bps with Sharpe 0.67–0.82 but degrades sharply at 50 bps to Sharpe 0.22, consistent with gross returns being only marginally above realistic execution costs at that level.

Measured weekly one-sided turnover (§7.6, Table 9) is 33.8% (Q1) and 25.7% (Q5) for baseline LMW. At 20 bps round-trip cost the combined LMW transaction cost is approximately 23.8 bps/week, or about 12 percentage points annualized — roughly 30% of the 39.5% gross return, leaving cost-adjusted Sharpe 0.67 and net annualized return 27.7% (Table 10). The vol-conditioned high-vol LMW has higher gross turnover (41.0% Q1 / 30.8% Q5) but markedly higher gross return (81.1%/yr), so its cost-adjusted Sharpe holds at 1.15 at 20 bps and at 0.77 even at 50 bps.

4.3 Regime Dependence

4.3.1 Event-anchored regimes

To facilitate comparison with the RP-PCA results and to build on compelling classifications, we adopt the event-anchored classification used in Kiefer and Nowotny (2026). Specifically we report LMW performance within four event-based regimes. CRISIS is anchored on the Terra/LUNA collapse (May 2022) and the FTX bankruptcy (November 2022); RECOVERY spans the crypto winter, ETF rally, and 2024 consolidation.

Regime	Date range	Days	N weeks
BULL	2021-01-01 → 2021-11-10	314	∼37
CRISIS	2021-11-11 → 2022-11-11	366	52
RECOVERY	2022-11-12 → 2024-11-05	725	104
NEW_BULL	2024-11-06 → 2025-12-31	421	60

Table 11: LMW Sharpe by event-anchored regime × formation ($K = 1$, NW t-stat in parentheses)

Regime	J=6w	J=8w	J=10w	J=12w
BULL	−0.31 (−0.25)	+1.87 (+1.71*)	−0.03 (−0.03)	+1.58 (+1.20)
CRISIS	+1.74 (+1.63)	+1.54 (+1.36)	**+2.44 (+2.29**)**	+1.16 (+0.97)
RECOVERY	+0.01 (+0.01)	+0.13 (+0.21)	+0.28 (+0.44)	−0.15 (−0.24)
NEW_BULL	+0.78 (+0.71)	+0.79 (+0.71)	+0.97 (+0.83)	+0.72 (+0.84)

LMW $J = 10$ in CRISIS earns 76% annualized with Sharpe 2.44 ($t = 2.29^{**}$); the result is unchanged after excluding LUNA, FTT, and other crisis tokens (§ Robustness). RECOVERY LMW is near zero across all J and NEW_BULL is statistically insignificant. These gaps are recovered by the subsequent vol-conditioning and mega-cap-exclusion sorts.

4.3.2 Vol conditioning by regime

The conditional vol double sort of §4.2 revealed that reversal is roughly six times stronger for high-volatility tokens over the full sample. We re-run the same sort within each event-anchored regime to test whether the vol channel operates continuously or concentrates in particular regimes.

Table 12: Vol-conditional LMW Sharpe by regime × formation ($K = 1$, NW t-stat)

Regime	Vol group	J=6w	J=8w	J=10w	J=12w
BULL	Low	+0.29 (+0.24)	+1.36 (+1.23)	+1.46 (+1.22)	+1.99 (+1.49)
BULL	**High**	−0.55 (−0.44)	+1.89 (+1.66*)	+0.60 (+0.47)	+0.49 (+0.32)
CRISIS	Low	+0.03 (+0.02)	+0.59 (+0.47)	+1.18 (+0.95)	+0.97 (+0.89)
CRISIS	**High**	+1.04 (+0.98)	+2.13 (+2.17**)	**+3.22 (+3.30***)**	+0.89 (+0.93)
RECOVERY	Low	−0.67 (−0.85)	−0.39 (−0.51)	−0.27 (−0.37)	−0.36 (−0.46)
RECOVERY	**High**	+0.41 (+0.62)	+0.78 (+1.33)	+0.47 (+0.87)	+0.29 (+0.47)
NEW_BULL	Low	+0.17 (+0.14)	+0.17 (+0.15)	−0.31 (−0.26)	−0.30 (−0.25)
NEW_BULL	**High**	+1.06 (+1.23)	+1.11 (+1.07)	+0.97 (+0.94)	+0.32 (+0.41)

High-vol LMW Sharpe exceeds low-vol LMW Sharpe in every regime at every formation horizon (except BULL $J = 6$w and $J = 10$w, which are noisy 33–37-week subsamples). High-vol LMW in CRISIS earns Sharpe 3.22 with $t = 3.30^{***}$ at J=10w. RECOVERY LMW is near zero unconditionally, but within RECOVERY the high-vol group produces positive Sharpes (0.29–0.78) while the low-vol group is consistently negative. The high-minus-low gap at $J = 10$ is 0.53 in BULL, 2.04 in CRISIS, 0.74 in RECOVERY, and 1.28 in NEW_BULL — the crisis-period gap is roughly four times the average.

4.3.3 Subsample variants

We split the sample further to diagnose strengths and weaknesses of the strategy.

Table 13: LMW Sharpe by subsample, baseline vs. vol-conditioned (J=8, $K = 1$; high-vol leg uses 2-bucket vol pre-sort, then quintile signal sort)

Subsample	Period	N wk	Baseline (q=5) SR (t)	High-vol (q=5) SR (t)
Full Sample	2021-01 → 2026-03	269	+0.96 (+2.10**)	**+1.36** (**+3.16*****)
First Half (H1)	2021-01 → 2023-07	125	+1.34 (+2.08**)	**+1.70** (**+2.84*****)
Second Half (H2)	2023-08 → 2026-03	135	+0.29 (+0.51)	**+1.03** (**+1.68***)
Bear 2022	2022-05 → 2022-12	∼35	+2.64 (+1.73*)	—
Recovery 2023	2023-01 → 2023-12	∼53	−0.52 (−0.50)	—
Bull 2024-25	2024-01 → 2025-12	∼104	−0.01 (−0.01)	—

4.3.4 H2 high-vol mid-cap filter

Table 14: H2 high-vol LMW Sharpe by mega-cap exclusion (J=8, skip= 1, $K = 1$; trailing 13-week vol, 2 buckets; NW t-stat in parens). Mega-caps removed in order BTC, ETH, BNB, XRP, SOL, ADA, DOGE, DOT.

N_{drop}	n_{syms}	H2 high-vol q=3	H2 high-vol q=5	H2 low-vol q=3	H2 low-vol q=5
0	70	+1.24 (+1.77*)	+1.03 (+1.68*)	−0.18 (−0.25)	+0.30 (+0.42)
4	66	**+1.42** (**+2.00****)	**+1.17** (**+2.01****)	−0.15 (−0.23)	−0.05 (−0.08)
5	65	+1.29 (+1.88*)	+1.14 (+1.92*)	−0.28 (−0.47)	−0.40 (−0.64)
6	64	+1.31 (+1.84*)	+1.18 (+1.98**)	−0.25 (−0.43)	−0.17 (−0.27)
7	63	+1.37 (+1.84*)	**+1.38** (**+2.17****)	−0.56 (−0.96)	−0.24 (−0.40)
8	62	+1.14 (+1.63)	+1.33 (+2.08**)	−0.35 (−0.57)	+0.03 (+0.04)

Removing the four largest mega-caps (BTC, ETH, BNB, XRP) strengthens the H2 high-vol tercile performance (Sharpe +1.42, $t = +2.00^{**}$) and the quintile cell to $t = +2.01^{**}$. The point estimate is stable across $N_{\text{drop}} \in \{4, 5, 6, 7\}$: tercile Sharpes range 1.29–1.42 with t-stats $\geq 1.84^{*}$, and quintile t-stats clear 1.96^{**} at every drop level from 4 onward. The H2 weakness in Table 13 is largely a mega-cap phenomenon: the reversal premium lives in the high-vol mid-cap tail throughout the second half of the sample.

4.4 Sensitivity to event-driven outliers

To ensure the most extreme single-asset events are not driving the reversal result, we re-run the $J = 8$ baseline and vol-conditioned LMW after sequentially dropping LUNA, FTT, and ARDR (the three tokens with notable price gaps in §2.1). The vol-high LMW Sharpe is insensitive across all four panels: 1.36 (full panel) → 1.37 (drop LUNA) → 1.37 (drop LUNA + FTT) → 1.38 (drop LUNA + FTT + ARDR), with NW t-statistics holding at 3.16–3.18. The high-vs-low vol gap actually widens monotonically (68.0% → 68.0% → 70.9% → 72.8% annualized), and the baseline LMW Sharpe rises from 0.96 to 1.09 after dropping LUNA + FTT before reverting toward 1.00 with ARDR also removed.

4.5 Autocorrelation Structure and Variance Ratio Tests

We implement the Lo and MacKinlay (1988) heteroskedasticity-robust variance ratio test. Under the random walk null, the variance of q-period returns equals q times the variance of one-period

returns:

$$\mathrm{VR}(q) = \frac{\mathrm{Var}(r_t + r_{t+1} + \cdots + r_{t+q-1})}{q \cdot \mathrm{Var}(r_t)}.$$

VR > 1 implies positive serial correlation (momentum); VR < 1 implies negative serial correlation (mean-reversion).

Table 15: Lo-MacKinlay Variance Ratio Tests

Period q	Mean VR	Median VR	% Assets VR < 1	Mean z-stat
2	1.058	1.050	27%	+0.03
4	1.041	1.018	47%	+0.01
8	1.066	1.030	44%	+0.01
16	**0.915**	**0.868**	**63**%	−0.02

The pattern matches the ACF structure: VR is slightly above 1 at short horizons and drops below 1 at $q = 16$. The median VR(16) = 0.868 means that 16-week return variance is only 87% of what the random walk predicts, with 63% of assets showing VR < 1. The cross-sectional pattern is economically consistent with the portfolio-sort evidence.

4.6 Additional Considerations

Sample size and statistical power. With 269 weeks, many specifications produce t-statistics in the 1.5–2.1 range. The consistency across 22 significant specifications lends collective weight, but individual parametric tests have limited power. A non-parametric circular block bootstrap shows the vol-high LMW Sharpe exceeds 1.0 in 82% of 10,000 replications, and β_{PC1} from the 5-factor RP-PCA spanning is negative in 100% of replications for both the baseline and vol-high LMW. The baseline LMW Sharpe at point estimate 0.96 remains borderline in the bootstrap (47% of replications exceed 1.0, 98% exceed zero). Using stationary bootstrap does not materially alter the circular bootstrap results.

Mega-cap concentration. While we know the signal is driven by equal rather than size weighting, we test how mega-caps drive the reversal premium directly by re-running the $J = 8$ baseline and vol-conditioned LMW after excluding the top 4, 5, 6, and 8 names by average market cap (BTC, ETH, BNB, XRP, then SOL, ADA, DOGE, DOT). We find that removing the largest names strengthens every cell. Vol-high LMW Sharpe rises monotonically from 1.36 (full) to 1.61 (drop-4) to 1.69 (drop-8) with NW t-statistics climbing from 3.16 to 3.80; baseline LMW Sharpe peaks at 1.23 ($t = 2.60^{***}$, drop-5) versus 0.96 at the full panel; the vol-low LMW correspondingly weakens (SR 0.30 → 0.12 at drop-5) so the high-minus-low gap widens from 68% to 88% annualized. Mega-caps are low-vol relative to the rest of the cross-section and their inclusion attenuates rather than generates the premium.

5 Conclusion and Future Work

Cross-sectional reversal appears in cryptocurrency returns at 6-12 week formation horizons, peaking at J=8 weeks with a Sharpe ratio near 1.0 and a Newey-West t-statistic of 2.10. The effect is concentrated among volatile, mid-capitalization tokens: conditioning on high volatility raises the Sharpe to 1.37 (t=3.19), and excluding the top 8 tokens by market cap raises it further to 1.69 (t=3.80). Both results clear the Harvey, Liu, and Zhu (2016) threshold of 3.0. A circular block bootstrap with 10,000 replications confirms the high-volatility result nonparametrically, with the

Sharpe ratio positive in every replication and its 5th percentile at 0.67. The vol-high LMW survives 50 basis points of round-trip transaction costs with a Sharpe of 0.77.

The reversal and momentum effects appear to involve different parts of the crypto cross-section. Value-weighted momentum at J=1-3 weeks, consistent with Liu, Tsyvinski, and Wu (2022), operates among larger, more liquid tokens. Equal-weighted reversal at J=8-12 weeks operates among smaller, higher-volatility tokens. Together they trace out a compressed version of the equity momentum life cycle at roughly one-quarter to one-sixth the timescale, though whether this reflects a single underlying mechanism operating at different speeds or two distinct phenomena sharing a horizon boundary remains an open question.

Several channels could generate the volatility-dependent reversal we document. Nagel (2012) argues that reversal compensates liquidity providers, with the compensation scaling with the cost of absorbing flow in volatile assets. Our vol-conditioning results are consistent with this prediction, and the CRISIS $\times$ high-vol cell (Sharpe 3.22, t=3.30) shows the effect at its most intense where liquidity provision would be most costly. A direct test of the Nagel channel is feasible in crypto markets: onchain data from Uniswap liquidity pools allow measurement of LP costs per asset, which can be mapped to the return series.

The ability of the LMW portfolios to price the cross-section is limited. We run the Fama-McBeth procedure and find the individual λ_{LMW} is insignificant in univariate and multivariate specifications. Point estimates of the LMW price of risk are economically plausible and correctly signed but statistically indistinguishable from zero. The role of LMW in contributing to priced risk in the cross section of cryptocurrency returns is an important subject for future research.

Cryptocurrencies with active treasury management programs could be selling into rallies, dampening momentum effects and generating reversal effects among less liquid tokens. We present preliminary evidence in support of this explanation with the category sorts showing reversal is not present among the subset of tokens that have no discretionary issuance or treasury management programs. Future work should go beyond the category sorts and obtain data on treasury, foundation and market maker allocations and sales to understand the mechanism in more detail.

A Appendix

A.1 Portfolio Construction

This appendix collects the formal definitions of the risk-adjustment, factor-model, and optimization tools used in the main text.

Crypto market regression The portfolio-level market-beta regression used to compute alphas is

$$r_t^{\text{LMW}} - r_f = \alpha + \beta(r_t^{\text{MKT}} - r_f) + \varepsilon_t,$$

where $r_f = 3.5\%/52.18$ is the weekly risk-free rate. The market return r_t^{MKT} is constructed as either the equal-weighted (EW) cross-sectional mean of asset returns or the value-weighted (VW) return with lagged market-cap weights

$$r_t^{\text{VW}} = \sum_i w_{i,t-1} r_{i,t}, \qquad w_{i,t-1} = \frac{\text{MCap}_{i,t-1}}{\sum_j \text{MCap}_{j,t-1}},$$

with $\text{MCap}_{i,t-1} = P_{i,t-1} \cdot S_{i,t-1}$ (price times circulating supply from CoinGecko). Alpha t-statistics use Newey-West HAC standard errors at 6 lags.

Spanning regression. For a reference portfolio set $\{F_k\}$ — either the five RP-PCA principal-component return series or the RP-PCA tangency / min-var pair — we estimate

$$r_t^{\mathrm{LMW}} = \alpha + \sum_k \beta_k F_{k,t} + \varepsilon_t.$$

The OLS intercept satisfies the identity

$$\alpha = \bar{r}^{\mathrm{LMW}} - \sum_k \beta_k \bar{F}_k,$$

which makes explicit that the spanning residual depends on both the realized means of the reference portfolios and the fitted loadings. In a single-factor spanning regression against a tangency portfolio, $\alpha = E[r^{\mathrm{LMW}}] - \beta \cdot E[r^{\mathrm{tang}}]$ — the identity invoked in §F.4a to explain the configuration sensitivity of α across the 15-cell tangency sweep.

Tangency Portfolio. For a K-factor panel with factor-space mean μ_f and covariance Σ_f, the tangency and minimum-variance factor-space weights are

$$w_{\mathrm{tang}} = \frac{\Sigma_f^{-1}\mu_f}{\mathbf{1}^\top \Sigma_f^{-1}\mu_f}, \qquad w_{\mathrm{mv}} = \frac{\Sigma_f^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma_f^{-1}\mathbf{1}}.$$

Asset-space portfolio weights $\omega_t \in \mathbb{R}^N$ at rebalance time t are recovered by composing the factor-space weights with the estimated loadings:

$$\omega_t = \Lambda_t \, w_{\mathrm{factor},t}.$$

In the rolling OOS pipeline used in Appendix F, Λ_t is re-estimated every 7 days on a 180-day trailing window; μ_f is estimated on a sweepable mean-window in $\{14, 28, 42, 56, 70, 84, 120, 180\}$ days and Σ_f on the full 180-day window.

A.2 Newey-West HAC Standard Errors

All t-statistics in the paper use Newey-West (1987) heteroskedasticity and autocorrelation consistent (HAC) standard errors with a Bartlett kernel and $L = 6$ lags. The HAC variance estimator is

$$\hat{\sigma}^2_{\mathrm{NW}} = \hat{\gamma}_0 + 2\sum_{j=1}^{L}\left(1 - \frac{j}{L+1}\right)\hat{\gamma}_j,$$

where $\hat{\gamma}_j = \frac{1}{T}\sum_{t=j+1}^{T}(r_t - \bar{r})(r_{t-j} - \bar{r})$ is the sample autocovariance at lag j and the term in parentheses is the Bartlett kernel weight. The t-statistic against the null $H_0 : E[r_t] = 0$ is $t = \bar{r}/\sqrt{\hat{\sigma}^2_{\mathrm{NW}}/T}$. Six lags is appropriate for weekly data where overlapping portfolio construction induces serial correlation up to approximately the holding period. For subsamples with fewer than 100 weeks we use $L = 3$.

A.3 Circular and stationary block bootstrap of LMW Sharpe

The Newey-West HAC t-statistics of Sections 3–7 rely on asymptotic approximations that can be tight at our sample length ($T = 260$ weeks). As a non-parametric complement we run a circular block bootstrap in the spirit of Politis and Romano (1994), analogous to the implementation used

for RP-PCA in the companion paper (Kiefer and Nowotny, 2026). We resample contiguous blocks of weekly LMW returns, wrap circularly to preserve sample length, compute the Sharpe ratio on each replication, and repeat 10,000 times at three block sizes (4, 8, 12 weeks) bracketing the autocorrelation range induced by the $J = 8$ formation window.

Table 16: Circular block bootstrap of LMW Sharpe — mean [5%, 95%]

Variant (all $J = 8$, $K = 1$)	block = 4w	block = 8w	block = 12w	$P(\mathrm{SR} > 0)$	$P(\mathrm{SR} > 1)$
Baseline LMW	+0.95 [+0.24, +1.62]	+0.96 [+0.23, +1.66]	+0.96 [+0.25, +1.63]	0.984	0.467
Vol-Low LMW	+0.30 [−0.54, +1.12]	+0.30 [−0.57, +1.15]	+0.30 [−0.57, +1.16]	0.715	0.089
Vol-High LMW	**+1.37 [+0.67, +2.05]**	**+1.38 [+0.67, +2.07]**	**+1.38 [+0.72, +2.03]**	**1.000**	**0.821**
Extr-Low LMW	+0.42 [−0.29, +1.11]	+0.41 [−0.25, +1.09]	+0.42 [−0.24, +1.09]	0.844	0.080
Extr-High LMW	+0.78 [+0.06, +1.42]	+0.79 [+0.06, +1.44]	+0.77 [+0.07, +1.40]	0.963	0.306
Disp-Low LMW	+0.44 [−0.25, +1.06]	+0.45 [−0.27, +1.09]	+0.43 [−0.30, +1.08]	0.858	0.077
Disp-High LMW	+1.18 [+0.41, +1.95]	+1.18 [+0.40, +1.94]	+1.18 [+0.47, +1.88]	0.994	0.659

Probability statements are for block = 8 weeks; values at 4w and 12w are within 0.01 in every row, confirming the LMW autocorrelation is short-range and the bootstrap CIs are insensitive to the block choice in the 4–12 week range. The headline high-vol result at $J = 8$ is the strongest cell: 10,000 of 10,000 bootstrap replications yield a positive Sharpe, and 82% exceed 1.0. The baseline LMW is supported by 98% positivity but only 47% above Sharpe = 1.0 — consistent with its point estimate of 0.96 lying at the boundary.

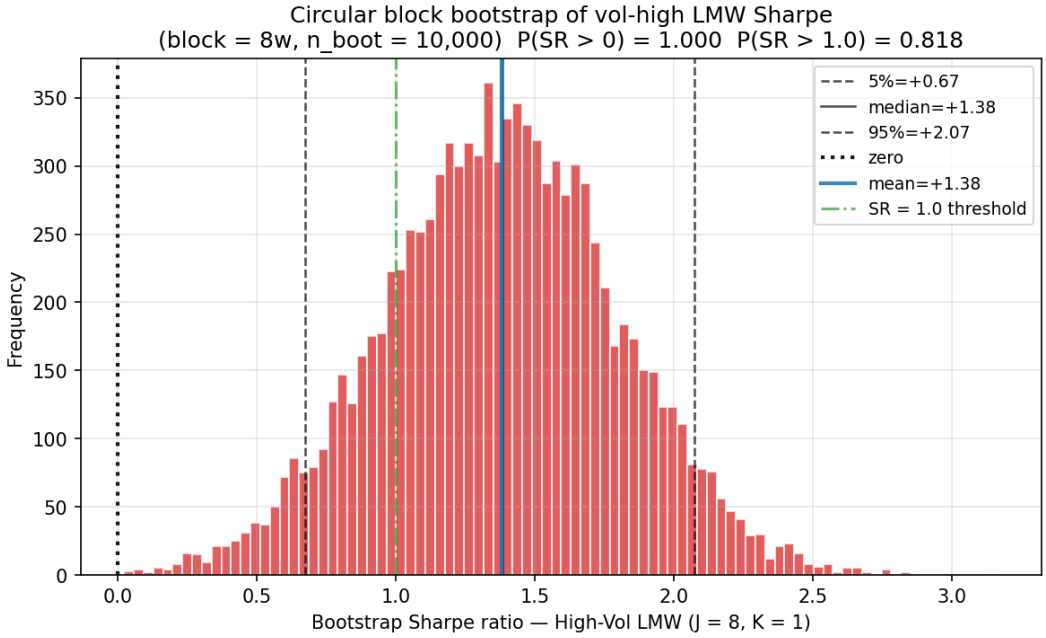

Figure 7: **Bootstrap distribution of the vol-high LMW Sharpe ratio.** Histogram of 10,000 circular-block-bootstrap Sharpe ratios for the vol-high LMW ($J = 8$, $K = 1$, block = 8 weeks). Vertical lines mark the 5th percentile (+0.67), median (+1.38), 95th percentile (+2.07), mean (+1.38), the zero line, and the SR = 1.0 threshold. The sampled bootstrap distribution lies above zero and 82% of replications exceed Sharpe = 1.0, consistent with the point estimate of 1.37 and Newey-West $t = 3.19$.

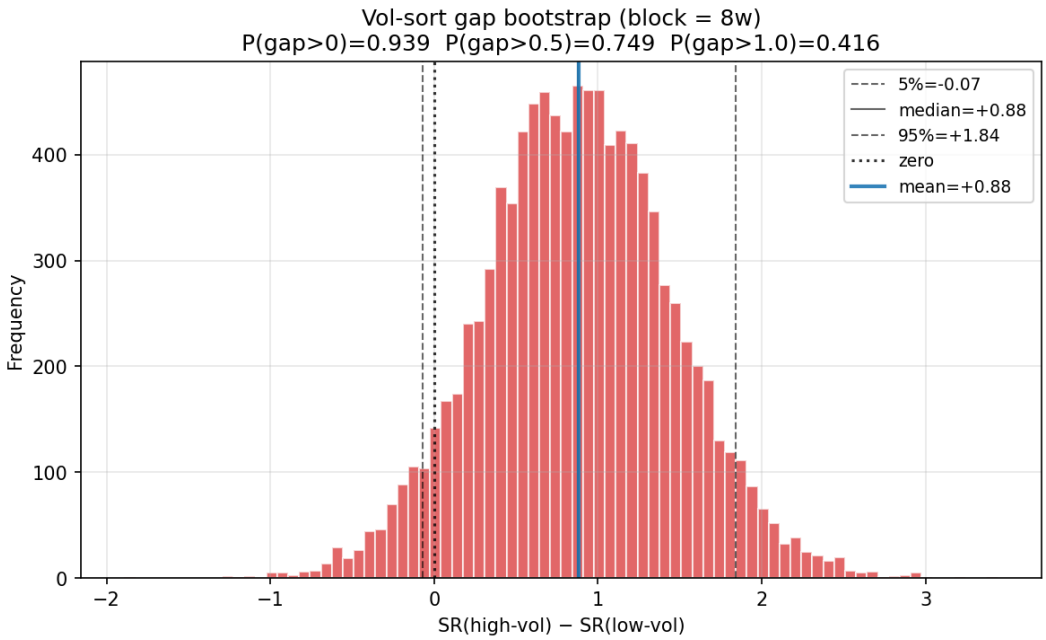

Figure 8: **Bootstrap distribution of the vol-sort Sharpe gap.** Histogram of $\mathrm{SR}_{\text{vol-high}} - \mathrm{SR}_{\text{vol-low}}$ across 10,000 joint bootstrap replications at block = 8 weeks. The gap is positive in 94% of replications, the 5th percentile lies at −0.07 (just below zero), and the 95th percentile at +1.84.

To confirm the bootstrap machinery is not understating tail width, we repeat the joint-panel exercise on a random 130-week contiguous subsample (2023-06-04 to 2025-11-23).

Table 17: Half-sample stress test — bootstrap CI and P(signal) at full $N = 244$ vs half $N = 130$

Statistic	Full $N = 244$	Half $N = 130$	CI-width ratio	Full P(signal)	Half P(signal)
$\mathrm{SR}_{\text{vol-high}} > 0$	$[+0.56, +2.05]$	$[-0.27, +1.80]$	**1.40**	0.9982	**0.8715**
$\mathrm{SR}_{\text{gap (high-low)}} > 0$	$[-0.07, +1.84]$	$[-0.34, +1.81]$	1.12	0.9385	0.8683
$\mathrm{SR}_{\text{gap (high-base)}} > 0$	$[-0.03, +0.82]$	$[-0.05, +1.13]$	1.40	0.9378	0.9289
$\beta_{\text{PC1, vol-high}} < 0$	$[-0.421, -0.247]$	$[-0.549, -0.218]$	1.89	1.0000	1.0000

The CI-width ratio for $\mathrm{SR}_{\text{vol-high}}$ is 1.40 — roughly the $\sqrt{2} \approx 1.41$ expected under correct calibration; $P(\mathrm{SR} > 0)$ drops from 0.998 to 0.872 as the sample halves, producing visible zero-crossings at the shorter sample. The full-sample 99.8% positivity is therefore a strong-signal consequence of the point estimate being far from zero.

β_{PC1} retains $P(< 0) = 1.000$ even at $N = 130$ because its point estimate is so far from zero (roughly 3.3 standard deviations at full sample, which remains 2.3+ standard deviations even at half sample). The β_{PC1} CI does widen by ratio 1.89 — slightly more than $\sqrt{2}$, consistent with β being a noisier statistic on a shorter panel — but not enough to produce tail crossings at 10,000 replications. The gap statistic widens least (ratio 1.12) because the contemporaneous high-low correlation the joint bootstrap preserves shrinks the SE of the difference.

Table 18: Joint bootstrap of SR(vol-high) − SR(vol-low) — the Nagel liquidity-provision gap

Block	Mean	5%	25%	75%	95%	$P(>0)$	$P(>0.5)$	$P(>1.0)$
4w	+0.88	−0.13	+0.42	+1.36	+1.88	0.926	0.736	0.420
8w	+0.88	−0.07	+0.42	+1.34	+1.84	0.939	0.749	0.416
12w	+0.88	−0.07	+0.43	+1.33	+1.83	0.936	0.743	0.414

Because the vol-high and vol-low LMW series share the same formation signal but are sorted into disjoint halves of the cross-section, we bootstrap them jointly (resampling the row index of the two-column panel in unison) to preserve the contemporaneous correlation. The gap is positive in 93–94% of bootstrap replications, positive by more than 0.5 in 74–75%, and positive by more than 1.0 (the paper's headline 6× multiple in crude units) in 41–42%. The 5% tail lies near zero, so a one-sided 5% test rejects the null of no vol-sort effect.

Vol-conditioning premium above the unconditional sort. A complementary metric is the Sharpe differential between the vol-high LMW and the unconditional baseline LMW. This isolates the contribution of vol conditioning beyond the plain quintile sort. Both variants share the same $J = 8$ formation signal and quintile construction, and differ only in whether the cross-section is first split into two vol halves before the signal sort. The point estimate is $1.37 - 0.96 = +0.41$.

Table 19: Joint bootstrap of SR(vol-high) − SR(baseline) — vol-conditioning premium

Block	Mean	5%	25%	75%	95%	$P(>0)$	$P(>0.25)$	$P(>0.5)$
4w	+0.39	−0.05	+0.19	+0.59	+0.85	0.925	0.698	0.353
8w	+0.39	−0.03	+0.19	+0.60	+0.82	0.938	0.717	0.335
12w	+0.39	+0.01	+0.20	+0.59	+0.79	**0.952**	0.721	0.322

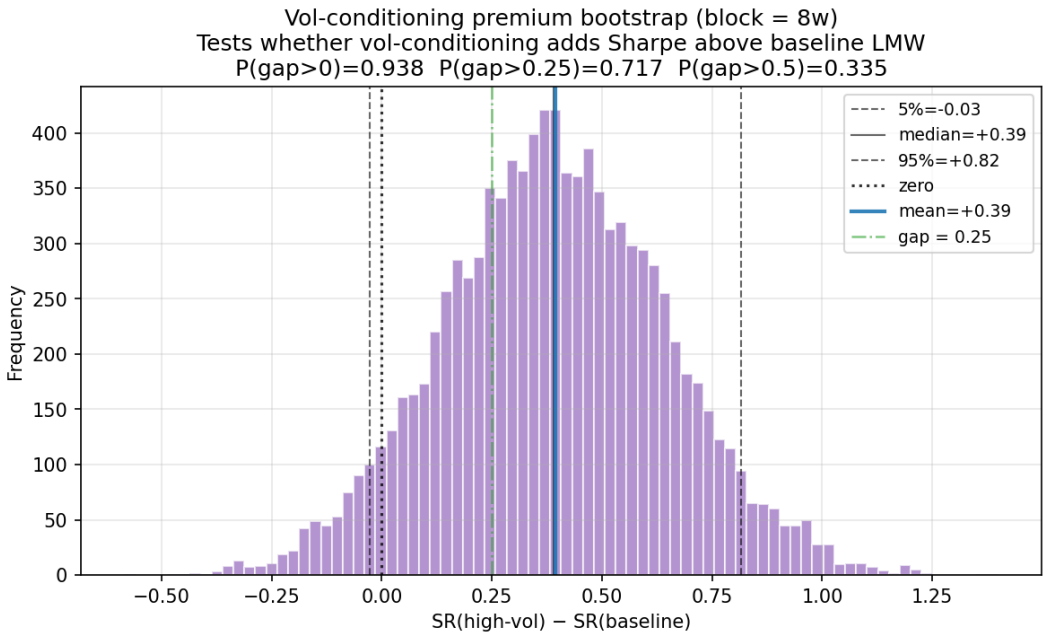

Figure 9: **Vol-conditioning premium over baseline.** Histogram of $\mathrm{SR}_{\text{vol-high}} - \mathrm{SR}_{\text{baseline}}$ across 10,000 joint bootstrap replications at block = 8 weeks. The gap is positive in 94% of replications, exceeds +0.25 (one-third of baseline LMW Sharpe) in 72%, and exceeds +0.5 in 34%. Although the high-minus-low gap (Table 18) is larger in magnitude, the high-minus-base gap is a tighter statistic. At block = 12 weeks the 5th percentile actually crosses above zero (+0.01), corresponding to a one-sided 5% rejection of no vol-conditioning premium.

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